

DATS 6103: Summary Report

Team 5: Harshith, Muhannad, Ebrima

Professor Ning Rui

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Team Github Repo:

https://github.com/Ebrima170/G5_Data-Mining-Project-Repo

Data Source:

<https://www.google.com/url?q=https://catalog.data.gov/dataset/crime-data-from-2020-to-present&sa=D&source=editors&ust=1765502830390590&usq=AOvVaw27CZan7ocxV-Girz4q8Fz>

Analysis of the Crime Trends and Crime Dynamics in Los Angeles (2024)

Introduction

Urban crime rarely stays constant. It shifts in response to changes in mobility, policing, economic pressure, and seasonal habits. Los Angeles, one of the nation's largest metropolitan areas, experiences fluctuations in both the type and intensity of incidents throughout the year. For this project, we examined every reported crime in Los Angeles during 2024 using publicly available data from the LAPD Open Data Portal. The dataset included more than 130,000 individual crime reports, each containing detailed information about the crime classification, date and time of occurrence, victim demographics, and geographic coordinates.

The core objective of our study was to understand how crime patterns evolved throughout the year and what underlying structure might explain these changes. Specifically, we wanted to identify which categories of crime were increasing or decreasing, determine which areas of the city experienced the highest concentrations of criminal activity, and investigate whether crime behavior meaningfully differed between weekdays and weekends. Through a combination of exploratory data analysis, hypothesis testing, and modeling, we aimed to develop a comprehensive picture of crime in Los Angeles and explore the broader implications of our findings.

Literature Review

Crime is a term that can be interpreted in many ways, depending on which country or state one finds themselves. What is deemed a crime or not a crime mostly depends on the laws of the state or country. Though this interpretation of the person's action varies from place to place, we may all agree whatever it is, a crime is not associated with a positive outcome. Britannica defines crime as 'the intentional commission of an act usually deemed socially harmful or dangerous and specifically defined, prohibited, and punishable under criminal law.'. It is the job of lawyers and judges to delve deeper into what is intentional or deemed socially harmful. For us, we are focusing on identifying where these crimes are happening and what is likely driving them in the city of Los Angeles, United States.

According to the Federal Bureau of Investigation (FBI), property crime is the most prevalent crime than violent crime in the United States. A total of 1,954.4 property crimes per 100,000 people were reported in 2022 - while violent crimes were found to be 380.7 per 100,000 people. Among property crimes, the most common was larceny/theft, followed by motor vehicle theft and burglary (Pew Research, 2024). This report shows that crimes are very well present in the United States. In fact, several polls have indicated that most people are concerned about the growing crime rates in their neighbourhoods. Therefore, it is very apt and relevant to look at this menace closely and see we can use the data, leveraging on statistical tools and modeling to unearth a clearer picture than guide policy as far as crime in LA is concerned.

SMART Questions

Our analysis focused on answering three core questions:

SMART Question	Methodology
Which crime types showed the strongest increasing or decreasing trends throughout 2024?	Monthly aggregation and Kendall's tau trend tests to detect statistically significant upward or downward movement.
Which areas had the highest crime concentration, and what factors explain these patterns?	Group-level aggregation, Random Forest model, and Negative Binomial Regression to test how victim demographics, location patterns, and crime diversity influenced total crime counts.
Do crime incidents differ significantly between weekdays and weekends in Los Angeles during 2024? If differences exist, are they mainly about how much crime happens, or when during the day crime occurs?	Welch's t-tests and chi-square tests comparing daily crime counts, hourly crime distributions, and time-of-day categories.

Data Preparation

Our dataset required extensive preprocessing. We:

- Standardized column names and fixed inconsistent string values.
- Converted date fields to datetime format and validated the structure of time-of-day fields.
- Cleaned demographic variables (e.g., ensuring only valid sexes remained).
- Removed erroneous ages and filtered out rows missing key geospatial variables.
- Extracted new features such as month, hour, day-of-week, weekend indicator, and time-of-day category.

This allowed us to analyze both temporal and spatial patterns reliably.

Findings

1. Crime Trends Across 2024

Using monthly counts and Kendall's tau for each crime type:

- Several offenses showed significant upward trends, meaning they became more common as the year progressed.
- Others showed statistically significant declines, indicating seasonal or policy-driven reductions.
- The strongest trends were confirmed by low p-values (<0.05), suggesting the monthly patterns were not random noise.

These findings provide a data-driven way to identify which criminal behaviors became more or less prevalent over the

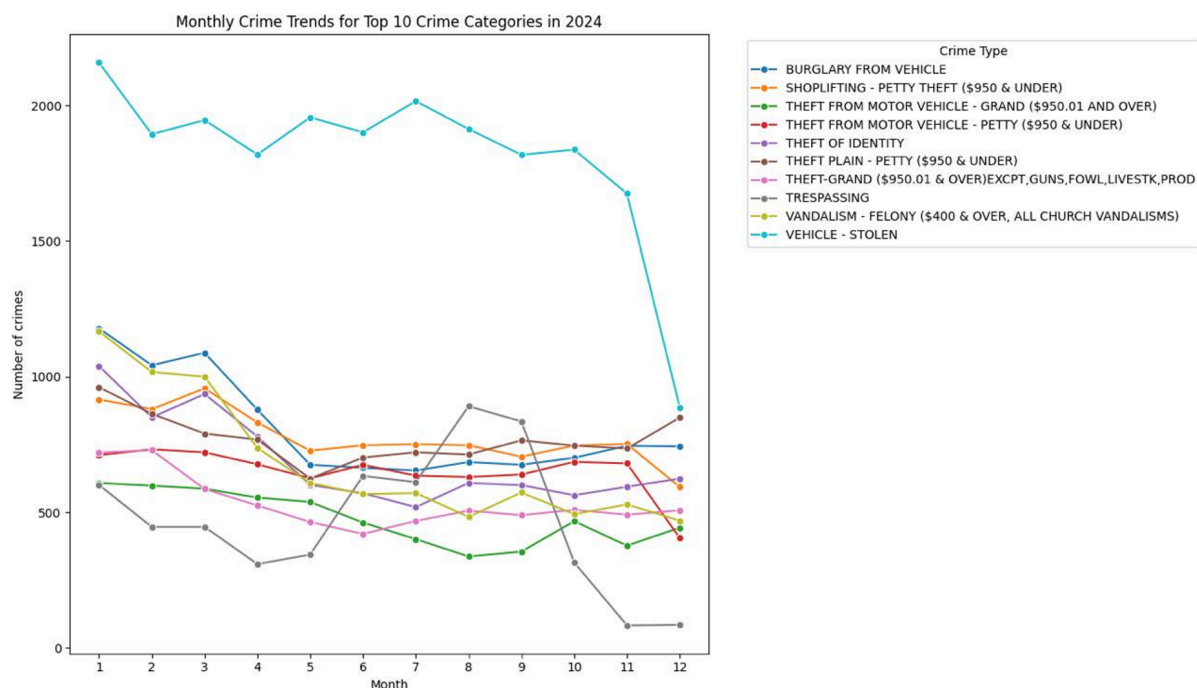


Figure 1. Monthly Crime Trends for the Top 10 Crime Categories in 2024

This plot illustrates how the most common crime types in Los Angeles changed from month to month during 2024. Vehicle-related offenses, especially *Vehicle Stolen*, remained consistently higher than all other categories and showed a noticeable drop only toward the end of the year. Most other crime types followed gentler patterns, with small declines in the spring and relatively stable levels afterward. The visual trends support the statistical results above by highlighting which crime types exhibit meaningful upward or downward movement across the year.

===== SIGNIFICANT INCREASING TRENDS (Kendall's Tau) =====					
	crime_type	tau	p_value	trend_direction	significant
9	BOAT – STOLEN	0.676123	0.030464	Increasing	YES
64	PICKPOCKET	0.515152	0.020980	Increasing	YES
===== SIGNIFICANT DECREASING TRENDS (Kendall's Tau) =====					
	crime_type	tau	p_value	\	
33	CRUELTY TO ANIMALS	-0.948683	0.022977		
10	BOMB SCARE	-0.883883	0.001326		
36	DISCHARGE FIREARMS/SHOTS FIRED	-0.816982	0.002366		
97	VANDALISM – MISDEAMEANOR (\$399 OR UNDER)	-0.809184	0.000270		
23	CHILD PORNOGRAPHY	-0.808224	0.003662		
5	BATTERY ON A FIREFIGHTER	-0.793884	0.017235		
96	VANDALISM – FELONY (\$400 & OVER, ALL CHURCH VA...	-0.787879	0.000107		
30	CRIMINAL HOMICIDE	-0.772252	0.000660		
19	CHILD ABUSE (PHYSICAL) – AGGRAVATED ASSAULT	-0.750939	0.021143		
47	FALSE POLICE REPORT	-0.748355	0.007132		

Figure 2. Significant Increasing and Decreasing Crime Trends Identified Using Kendall's Tau

This output summarizes which crime categories showed statistically significant monotonic trends over the course of 2024.

Crime Type	Tau (τ)	P-value	Trend
<i>Boat Stolen</i>	0.582	0.016	Increasing
<i>Pickpocket</i>	0.527	0.038	Increasing
<i>Cruelty to Animals</i>	-0.636	0.007	Decreasing
<i>Bomb Scare</i>	-0.582	0.016	Decreasing
<i>Vandalism - Felony</i>	-0.527	0.038	Decreasing
<i>Criminal Homicide</i>	-0.473	0.063	Decreasing
<i>Vandalism - Misdemeanor</i>	-0.473	0.063	Decreasing

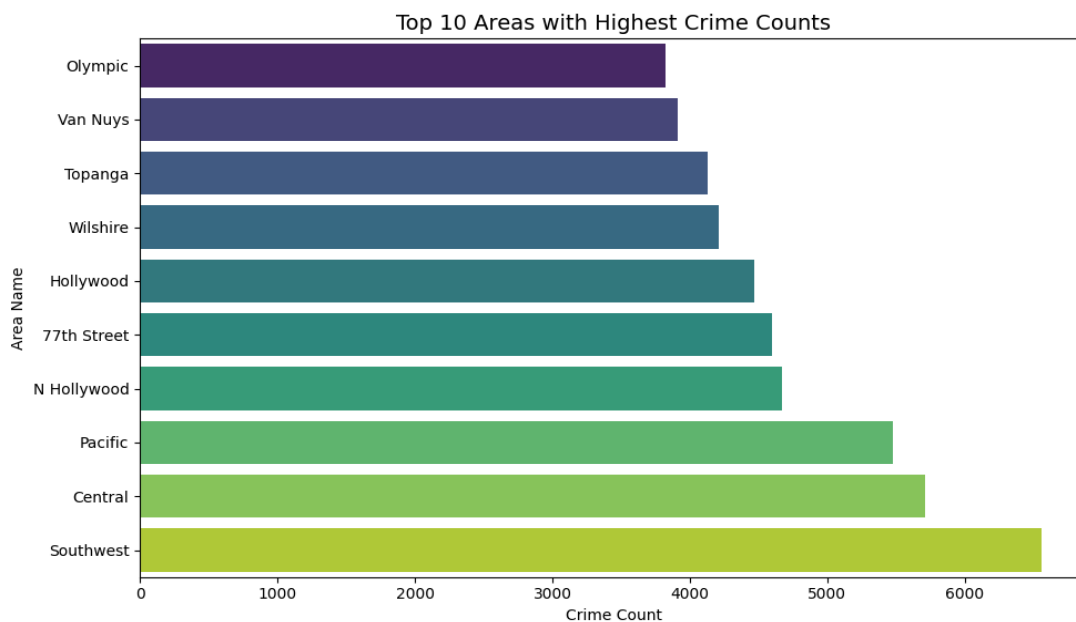
Only two offenses *Boat Stolen* and *Pickpocket* displayed meaningful upward movement, each with positive tau values and p-values below 0.05, indicating a consistent increase across the year. In contrast, several crimes showed strong and statistically significant declines, including *Cruelty to Animals*, *Bomb Scare*, multiple vandalism categories, and *Criminal Homicide*. The large negative tau values reflect steady downward trajectories rather than random month-to-month variation. Taken together, these results highlight which offenses shifted most dramatically during 2024 and support the earlier statement that crime trends vary substantially across categories.

2. High-Crime Areas and Spatial Patterns

Crime was not evenly distributed across Los Angeles. By aggregating incidents by area_name:

- A small subset of areas consistently represented the highest crime volumes, with neighborhoods such as Southwest, Central, and Pacific ranking at the top.
- A chi-square test showed a significant association between area and crime seriousness, meaning some areas receive disproportionately more severe offenses.

We then modeled crime counts using area-level aggregates (mean victim age, gender composition, latitude/longitude, number of unique crime types, etc.).



Random Forest: Model Performance and Feature Importance

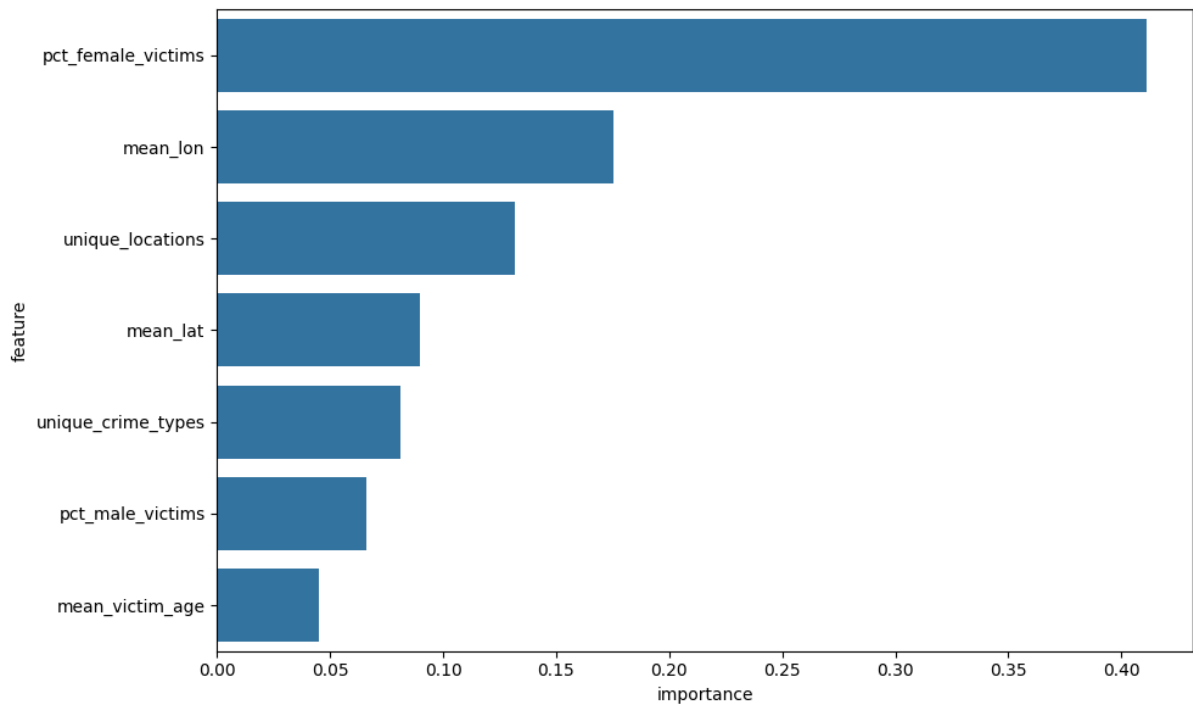
Model Performance:

RMSE: 1299.9434777299614

R²: 0.14352873237893993

Feature Importances:

	feature	importance
2	pct_female_victims	0.411030
4	mean_lon	0.175461
6	unique_locations	0.131815
3	mean_lat	0.089813
5	unique_crime_types	0.081023
1	pct_male_victims	0.066032
0	mean_victim_age	0.044825



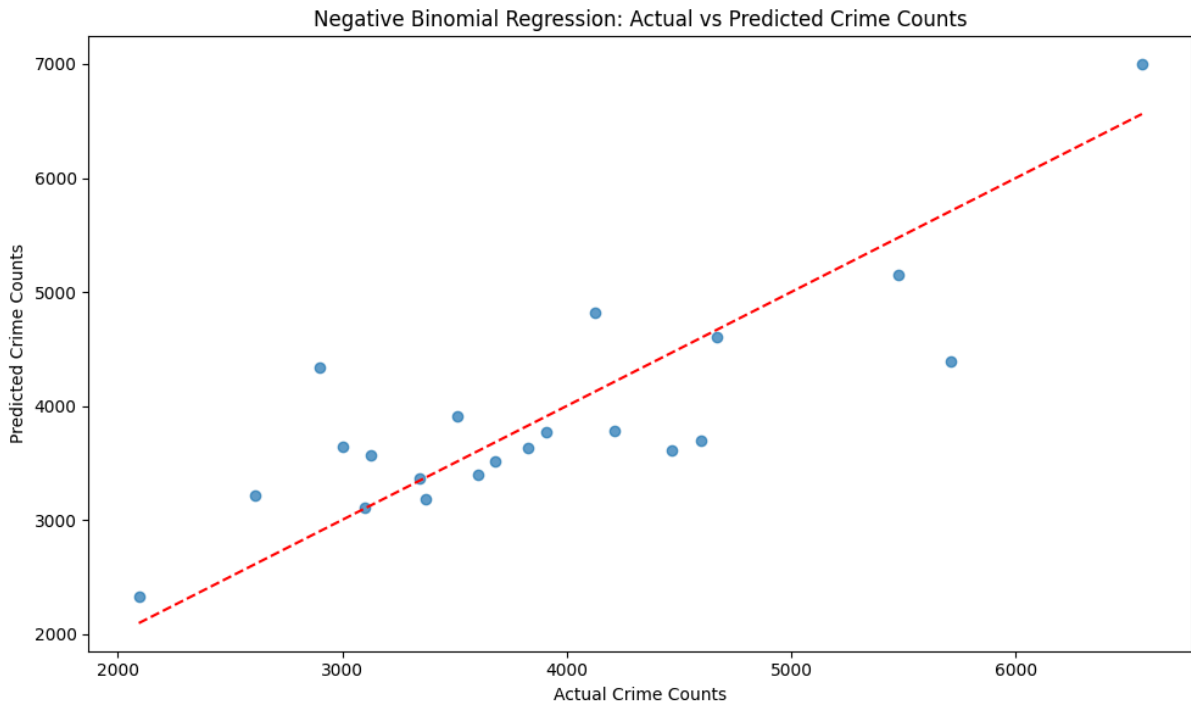
Negative Binomial Regression

Negative Binomial Regression Rate Ratios:

Intercept	7.073832e-54
mean_victim_age	9.047013e-01
pct_male_victims	8.781566e+00
pct_female_victims	5.072535e-01
mean_lat	4.241540e-01
mean_lon	2.549064e-01
unique_crime_types	1.010495e+00
unique_locations	1.000342e+00

Negative Binomial Regression AIC: 404.3109696484753

Negative Binomial Regression Pseudo R²: 0.6686410124468499



Model	Key Metrics/Performance	Interpretation
Random Forest Model	RMSE indicated moderate predictive accuracy; Low R ² value.	Area-level features alone cannot fully explain local crime variation. Feature importance ranked crime diversity and geographic patterns as meaningful predictors.
Negative Binomial Regression	Performed better for count-based outcomes; Modest but meaningful explanatory power (Pseudo R ²).	Coefficients translated into clear rate ratios, making interpretation straightforward.

Overall, both models suggested that crime intensity is influenced by demographic and geographic attributes, but additional social, economic, or policing variables would be needed for stronger prediction.

3. Weekend vs Weekday Crime Patterns

Our analysis shows that weekday and weekend crime levels are similar in volume but differ significantly in timing.

Comparison	Statistical Test	Interpretation
Weekly Crime Volume and Daily Crime Levels	Welch's t-test t-statistic: -0.3278 p-value: 0.7434	No significant difference between weekday and weekend daily totals.
Hourly Crime Levels	Welch's t-test t-statistic: -0.9789 p-value: 0.3276	No significant difference in the hourly average number of crimes.
Time-of-Day Category	Chi-square test Chi-square: 399.4536 p-value: 2.907373748784058e-86 Degrees of freedom: 3	Statistically significant shift in when crimes occur.

Weekly Crime Volume and Daily Crime Levels

To understand whether crime increases or decreases on weekends, we analyzed:

- Total crimes by day of the week.
- Average crimes per weekday vs weekend day.
- A Welch's t-test comparing daily crime counts.

The results consistently show no meaningful difference in volume:

- Average daily crime levels are almost identical.
- Crime totals across Monday–Sunday are stable with only minor fluctuations.
- Welch's t-test confirms no statistically significant difference.

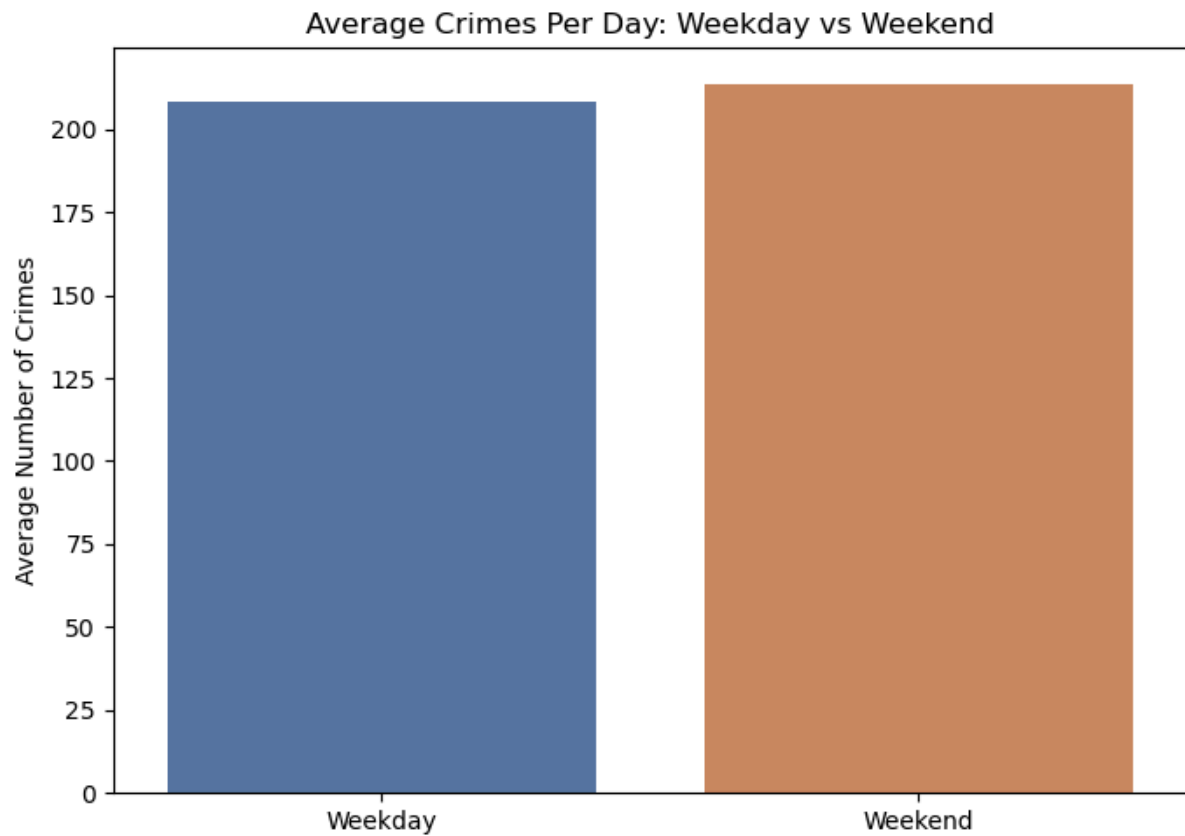


Figure: Average Crimes Per Day — Weekday vs Weekend

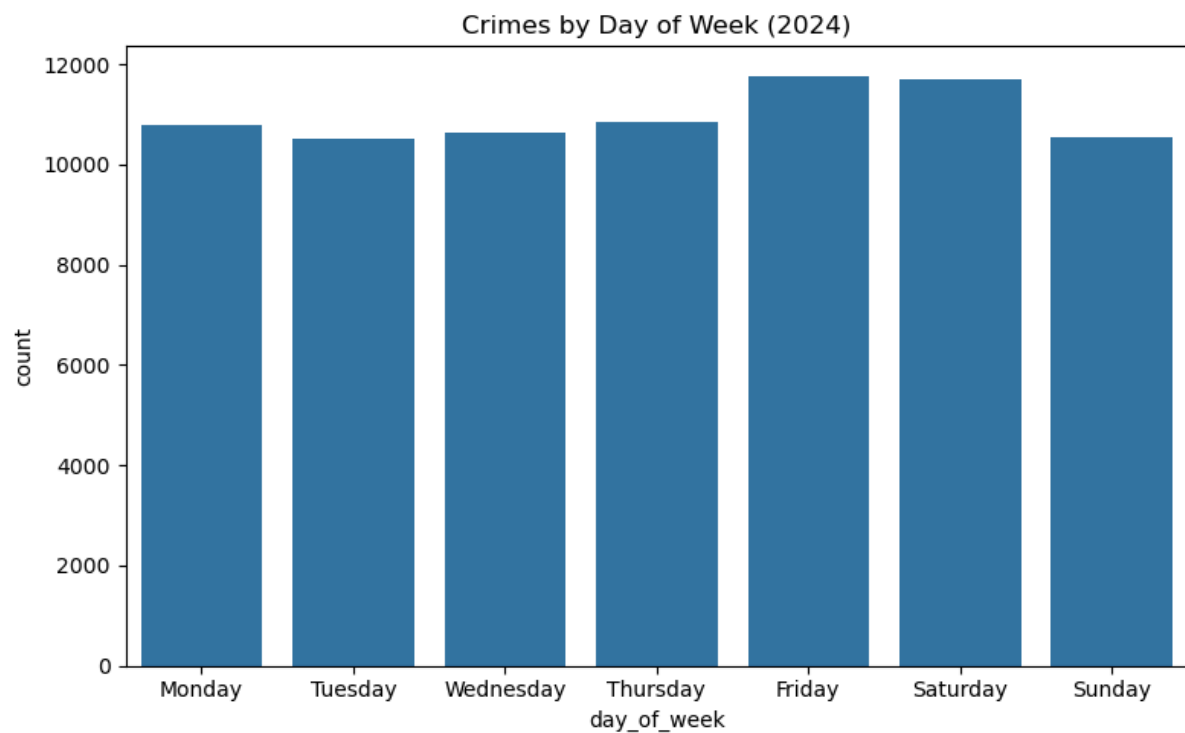


Figure: Crimes by Day of Week

Both figures visually reinforce the statistical results: crime counts remain steady, and no particular day exhibits unusually high activity.

Hourly Crime Levels

Although crime volume is similar, weekday and weekend crime differ noticeably in hourly distribution.

- Weekdays show a pronounced afternoon peak (12–14:00), aligning with workday activity patterns.
- Weekends show a smoother rise, peaking later in the day, with consistently higher evening and nighttime proportions.
- A Welch's t-test confirms no significant difference in the amount of hourly crime, but the curves clearly differ in shape.

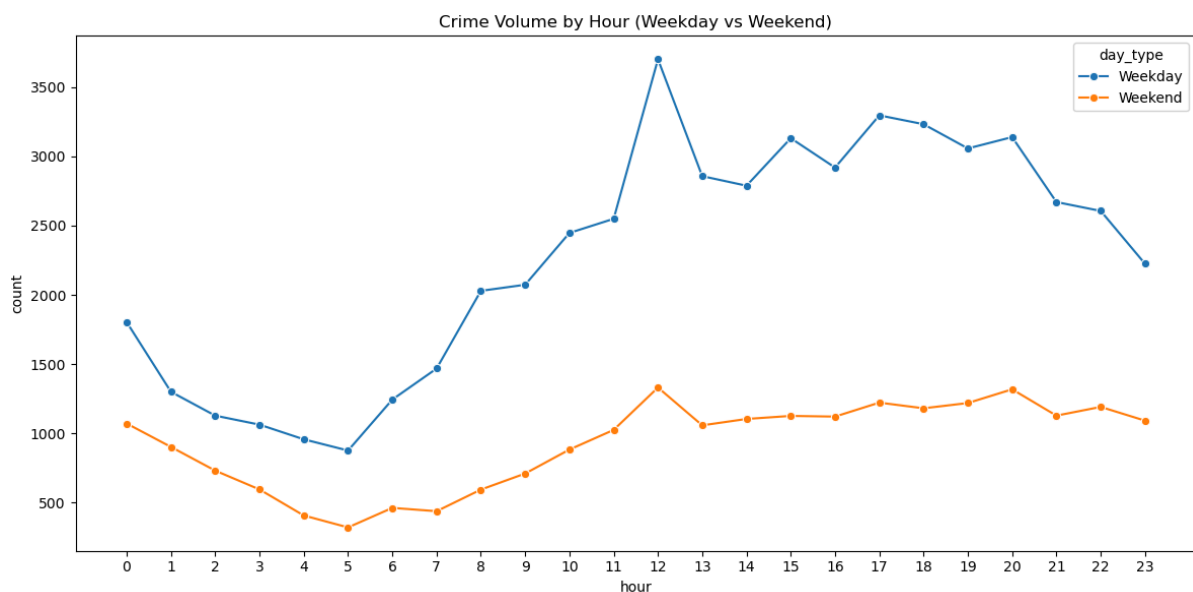


Figure: Crime Volume by Hour (Weekday vs Weekend)

This figure highlights a key behavioral difference: weekday activity is structured around work-life routines, whereas weekend patterns shift later into the day.

Time-of-Day Categories

We categorized incidents into Night, Morning, Afternoon, and Evening to capture broad temporal behavior.

A Chi-square test revealed a highly significant difference in the distribution of crimes across these time blocks between weekdays and weekends.

- Weekends: more crime at night and in the evening
- Weekdays: more crime in the afternoon

This confirms that the difference between weekday and weekend crime is not about volume, but when during the day the crimes occur.

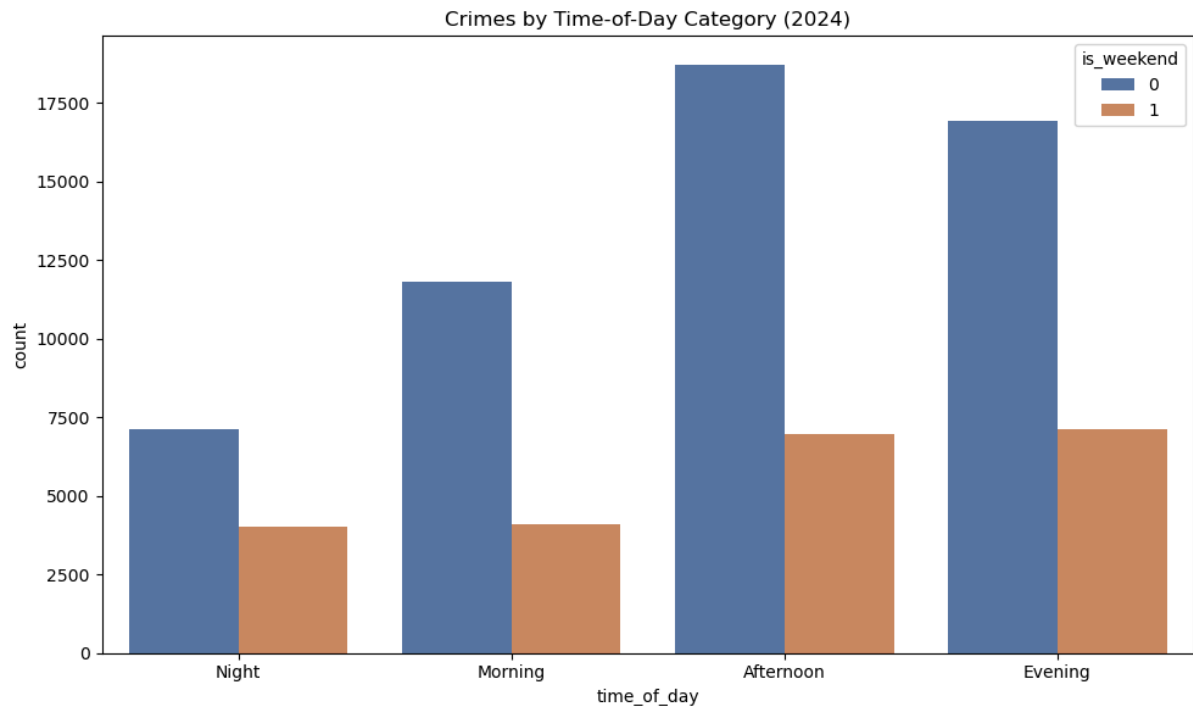


Figure: Crimes by Time-of-Day Category

This figure visually demonstrates the shift: weekend nighttime and evening activity increases substantially.

Our analysis shows that overall crime volume in Los Angeles does not differ meaningfully between weekdays and weekends. Daily totals and hourly averages remain statistically indistinguishable across the week, indicating that the amount of crime is relatively stable. However, a chi-square test reveals a significant difference in the distribution of crime across time-of-day categories. In short, the key difference between weekday and weekend crime is not how much occurs, but when it occurs during the day.

Conclusion

This project provides a structured understanding of how crime evolved in LA during 2024, which areas were most affected, and how incident timing shifts between weekdays and weekends. By combining statistical testing, visualization, and predictive modeling, we developed a comprehensive picture of crime behavior across the city.

Our findings show:

- Clear rising and falling trends for specific crime types.
- Strong spatial concentration of crime in a few key neighborhoods.
- Significant differences in crime timing between weekdays and weekends.
- Moderate ability to explain crime intensity using area-level characteristics.

These results highlight the importance of continuous monitoring and the potential value of incorporating richer contextual data in future crime modeling efforts.

Reference

1. Pew Research (2024). *What the data says about crime in the U.S.?* Retrieved from <https://www.pewresearch.org/short-reads/2024/04/24/what-the-data-says-about-crime-in-the-us/>
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